Skinasensyl™

New generation cosmeceutical ingredient for sensitive skin

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Market Data

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Sensitive Skin

Approximately 40% of the population, of all categories and phototypes, complains of having sensitive skin: they experience sensations of discomfort, tingling, burning and intolerance to certain types of product.
Sensitive skin knows no age or skin type.

Definitions of sensitive skin:

- **Excessive reaction of skin defense against stress** which normally would cause no reaction. **The threshold of skin tolerance to stress is decreased**.

- **Subjective skin hyper-reactivity to environmental factors or stimuli**. Consumers perceiving their skin as sensitive report exaggerated reactions, with subjective symptoms such as itching, burning, stinging or pain sensations.

Jackson EM: The science of Cosmetics. Am J Contact Dermat, 4: 108-10, 1993
Primavera G, Beradesca E: Sensitive skin: mechanisms and diagnosis. Int J Cosmet Sci, 27:1-10, 2005
Pons-Guiraud A: Sensitive skin: a complex and multifactorial syndrome. J Cosmet Dermatol, 3:145-8, 2004

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Sensitive Skin

Sensitive skin will react faster and more intensively to several parameters:

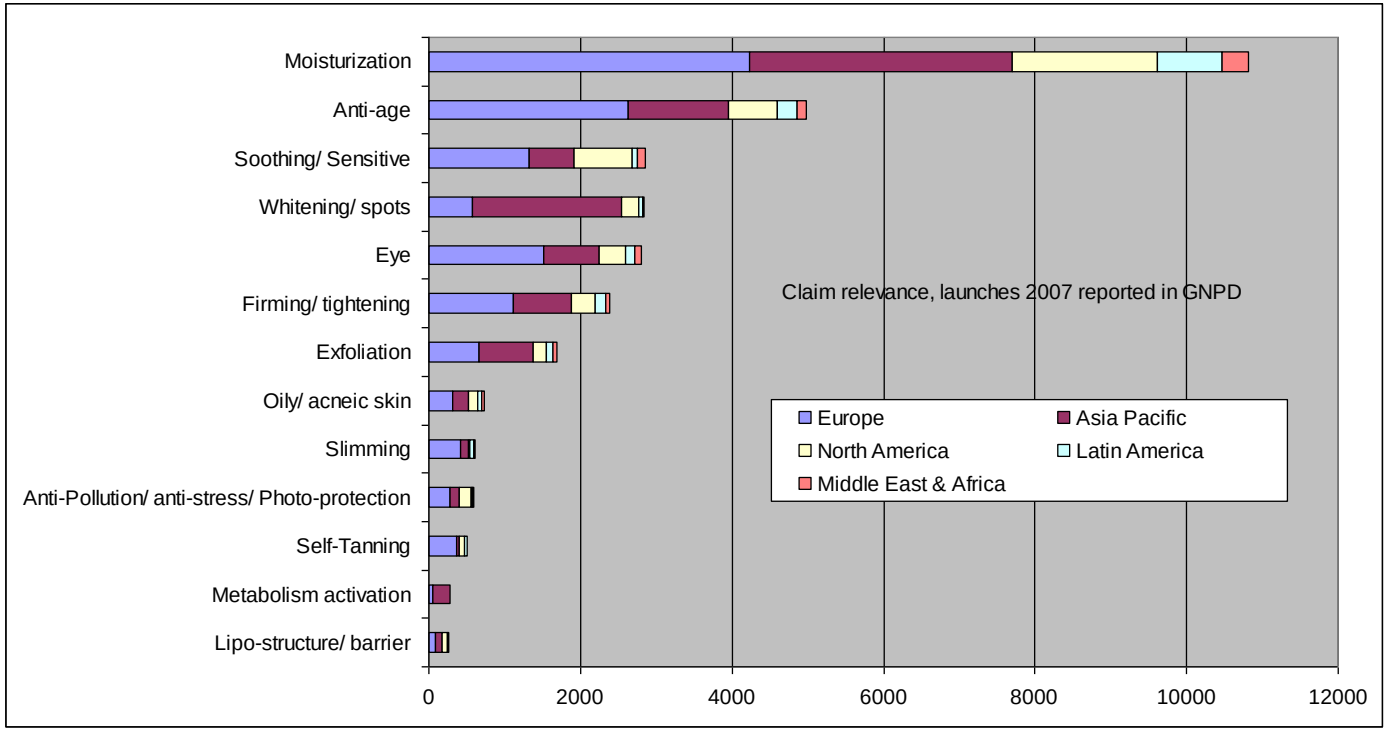
- **Outside factors**: temperature changes, heat as well as cold, sun…
- **The use of cosmetics products**: an increasingly important phenomenon, especially due to new type of products such as “home-use” treatments (dermabrasion and peels) as well as cosmeceutical products.
- **The use of certain medicines**.

- **Lifestyle** is also involved in the phenomenon of sensitive skin:
  - - stress, fatigue, emotions…
  - - tobacco, alcohol…

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Sensitive skin care: Market Size

Products for sensitive skin currently constitute one of the largest and one of the fastest growing segments of the skin care market.

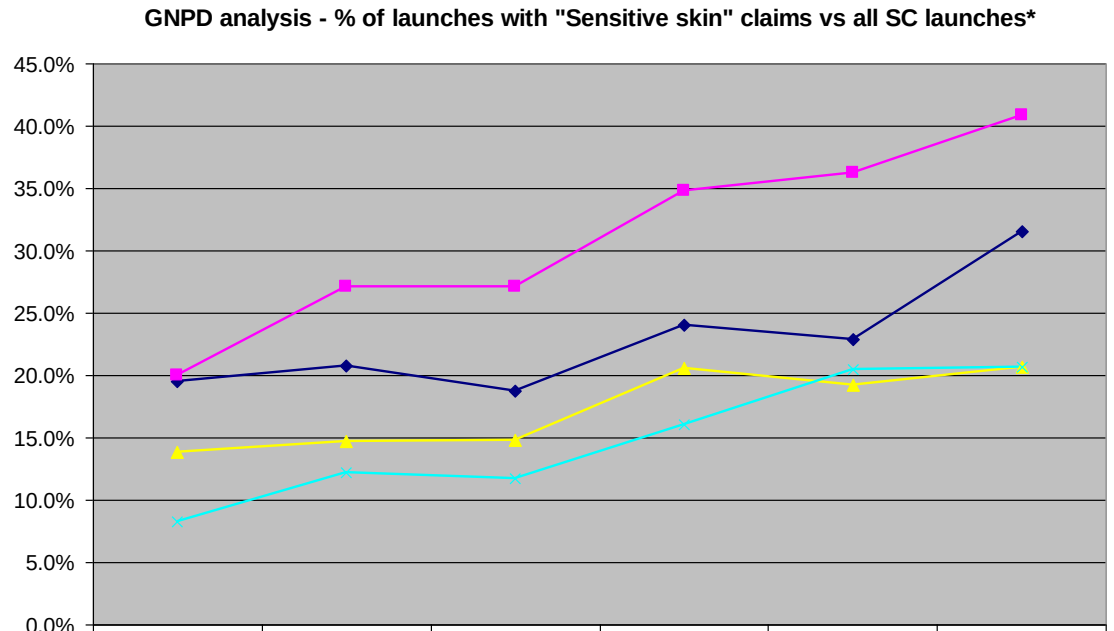


Search of launches containing at least one of key words "**anti-inflammatory" or "inflammation" or "irritation" or "irritations" or "soothing" or "soothe" or "soothes" or "redness" or "sensitive**".

Above mentioned Key Words are found in **24.2%** of total launches reported in GNPD in 2007.

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Products for Sensitive Skin: Market Evolution



	2003	2004	2005	2006	2007	2008, %
Europe	19.5%	20.7%	18.8%	24.1%	22.9%	31.5%
North America	20.0%	27.1%	27.1%	34.8%	36.2%	40.8%
Asia Pacific	13.9%	14.7%	14.8%	20.6%	19.3%	20.7%
Latin America	8.2%	12.2%	11.8%	16.1%	20.5%	20.7%

Growth of launches with "sensitive skin" claims over the last five years.

The growth is observed in all regions, but especially important in Europe and North America.

Following market mega trends, evolution of "sensitive skin" claim from specific "niche" ranges to add value to more generalist ranges.

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Products For Sensitive Skin: Market Drivers

- **Aging society**
  As people age, the skin's barrier function is compromised, making it more prone to water loss and outside aggressions. This can result in flaky and itchy skin, especially in cold, dry, windy climates, and in skin over-sensitivity.

- **Growing number of people with specific dermatological conditions**
  In between dermatological treatments, a cosmetic approach can also be considered.

- **Growing number of those who have had non-invasive surgery**
  'Aggressive' treatments such as Microdermabrasion, skin peels and laser resurfacing may lead to skin sensitivities and improper healing.

- **Growing market segments offering soothing claims as side benefits; segmentation for sensitive/sensitized skin**
  Men care (post shave products +6%*, men skin care +13.4%), sun care (sun protection: +10.4%*, after sun +5.4%). Other segments with "sensitive skin" claims: post depilation and baby care.

*Growth 2007 vs 2006, based on retails sales in Mln Eur

Post Dermatological Procedures Treatments

Darphin Intralderm has launched a Soothing Gel, which supplies a cooling sensation to sensitised skin. It is said to soothe redness and discomfort after a *procedure* such as a chemical peel.



Soothing Gel
Company: Darphin
Brand: Darphin Intralderm
Country: Ireland

Estée Lauder Advanced Night Repair Concentrate Recovery Boosting Treatment is described as a super serum for visibly damaged skin. The product also soothes and calms skin irritations and redness, and helps the skin recover from sun damage and *cosmetic procedures* such as chemical peels and micro-dermabrasions, leaving skin softer, and more supple.



Concentrate Recovery Boosting Treatment
Company: Estée Lauder
Brand: Estée Lauder Advanced Night Repair
Country: Spain

Source: GNPD

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Claims: Reduces Skin Sensitivity & Reactivity

"Reduce skin sensitivity."



Light Moisturizing Cream for Sensitive Skin
Company: La Roche-Posay
Brand: La Roche-Posay Hydreane
Country: Canada

"Product is claimed to decrease the cutaneous irritability … and restore skin comfort."



Hydra-Soothing Cream
Company: Laboratoires Dermatologiques Uriage, France
Brand: Uriage Eau Thermale Toléderm
Country: Singapore

Source: GNPD

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Concept

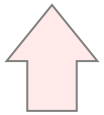
- New generation cosmeceutical ingredient
- Increase of skin tolerance threshold
- The neurocosmeceutical soother for over-reactive skin

Sensitive Skin: 3 Different Types

- Type I: **Low barrier function group**
  Alteration of skin barrier function with high TEWL and abnormal desquamation.

- Type II: **Inflammation group**
  Presence of inflammation, but normal barrier function.

- Type III: **Neurosensitive group**
  Normal barrier function, no inflammation, but exaggeration of the nerve response in skin (itching, discomfort, pain).



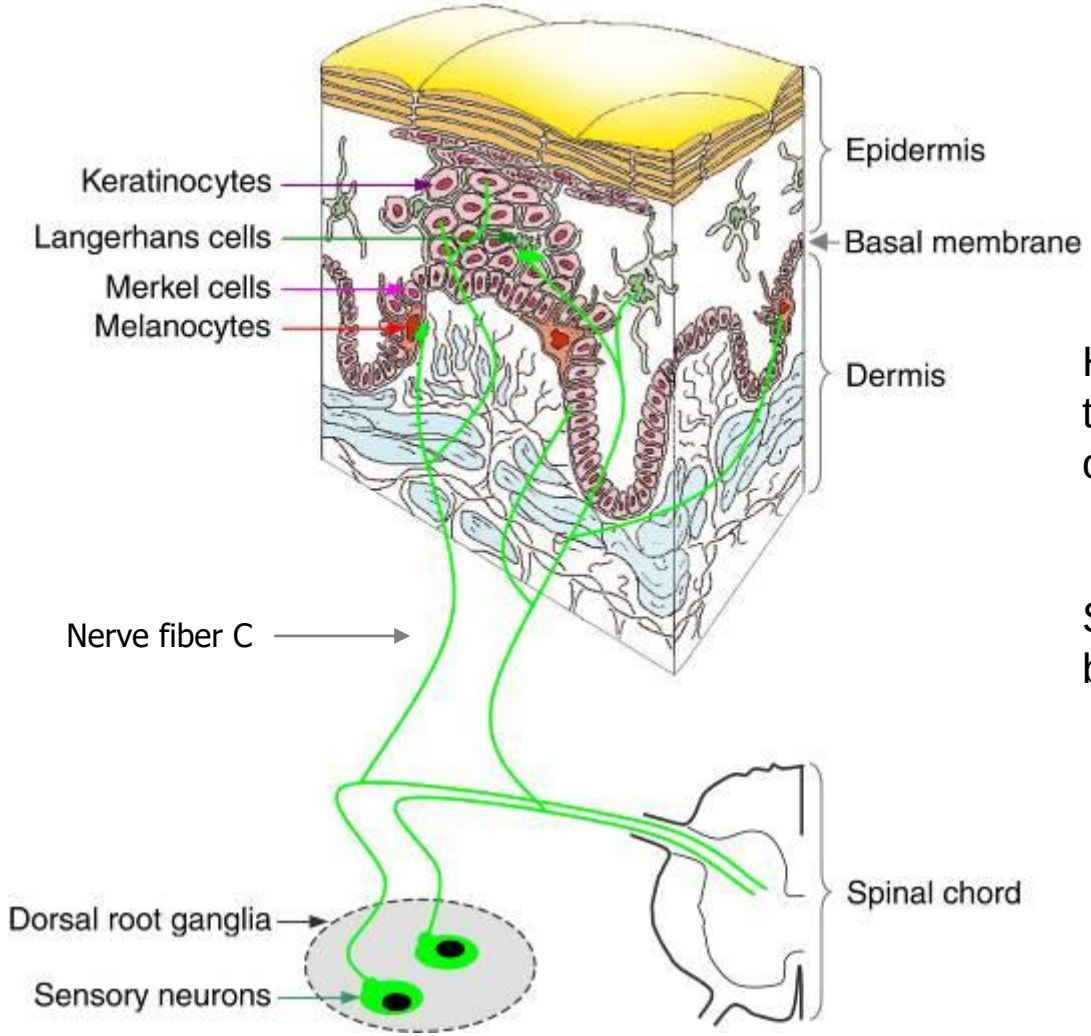
Our concept: **to decrease the nerve stimulation**

To decrease skin reactivity to different stimuli and to diminish sensations of pain and discomfort.

Yokota T *et al.*: Classification of sensitive skin and development of a treatment system appropriate for each group. IFSCC Mag, 6, 303-7, 2003

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Interactions Between Sensory Neurons And Skin Cells

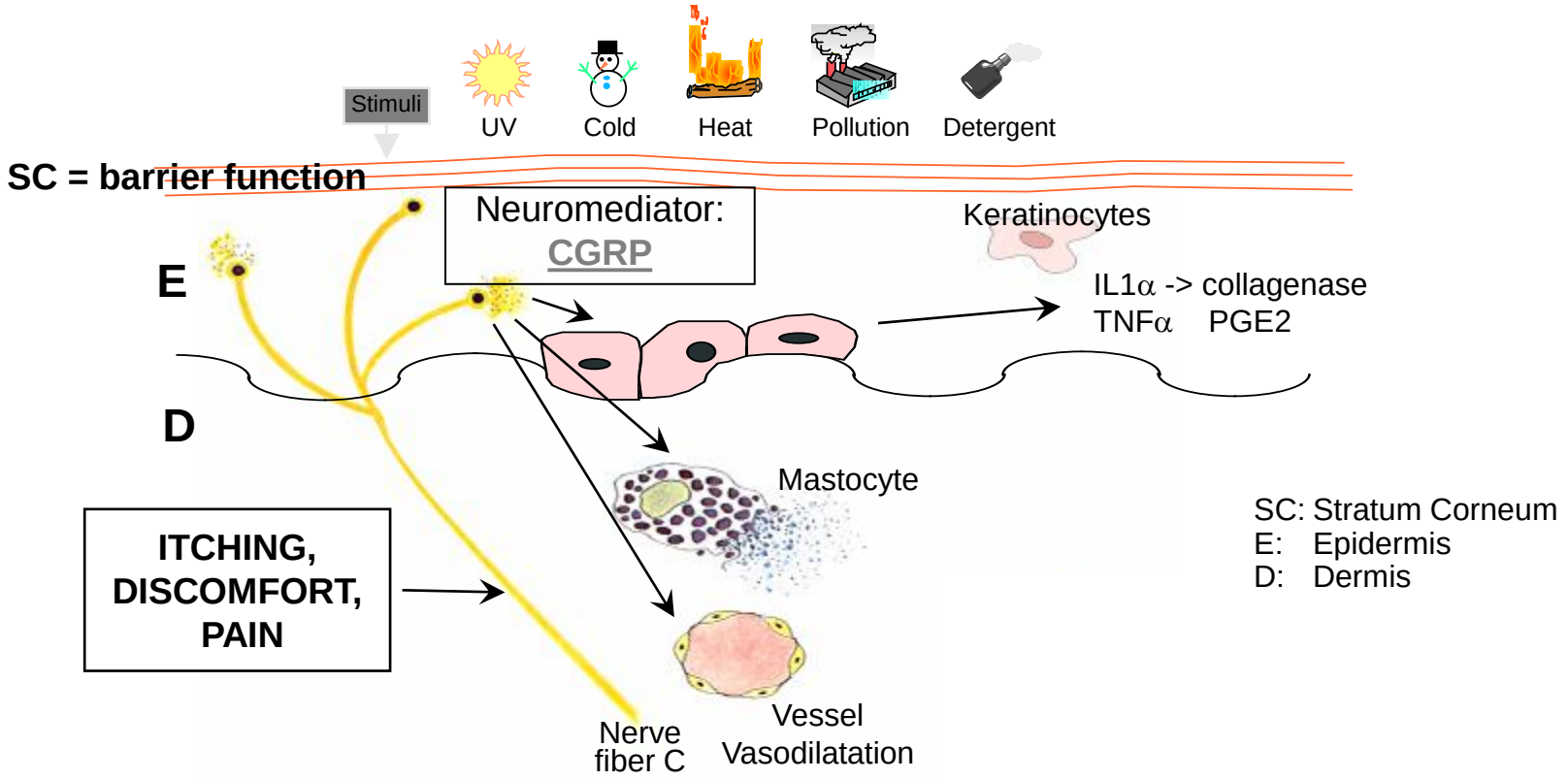


Human skin involves many nerve fibers which allow the central nervous system to be aware of the state of the skin and react to environmental stimuli.

Sensory nerve endings transmit pain, itchiness, burning or stinging sensations.

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Activation Of Peripheral Sensory Nerve Endings



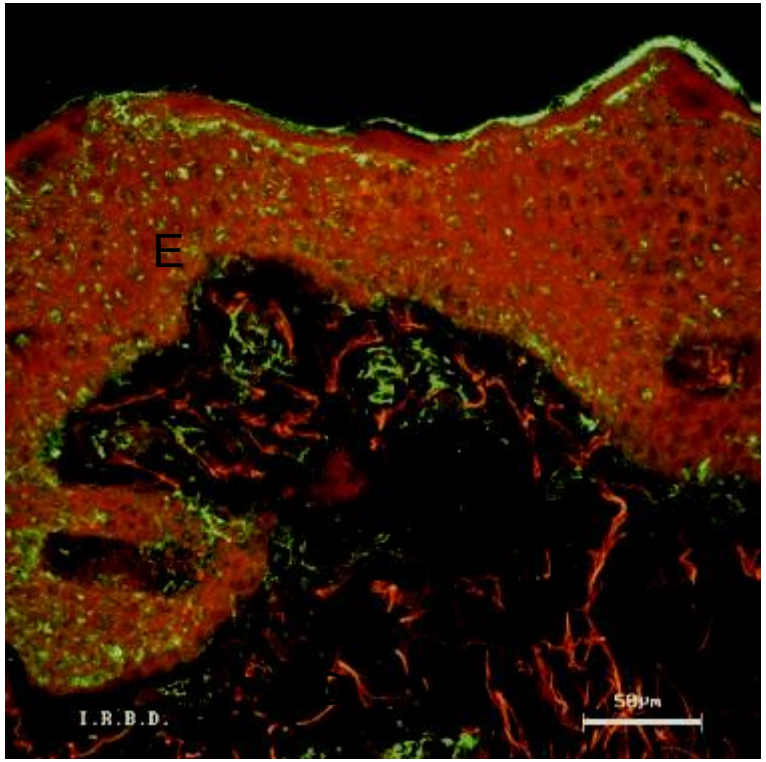
Stimulation of sensory nerve endings leads to:

- release of neuropeptides from nerve endings: pro-inflammatory messages, vasodilatation (local action),
- transmission of pain, discomfort, itching to the spinal cord (central action).

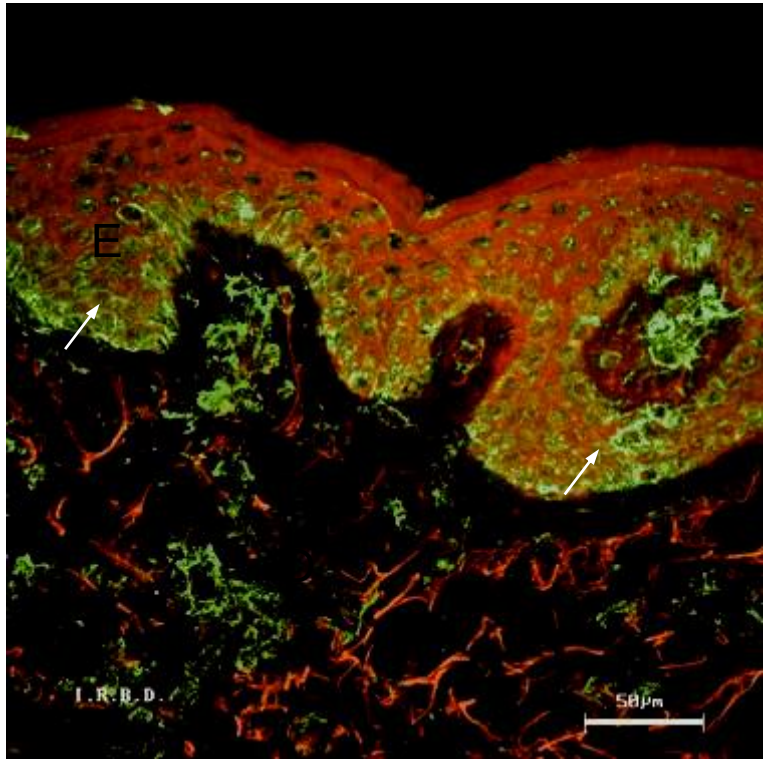
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Visualization Of CGRP in Human Skin (In Vivo)

Control



With stimulus (UV irradiation)



E = epidermis D = dermis

↗ CGRP in epidermis

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Cosmeceutical pure tetrapeptide for over-reactive skin

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Skinasensyl™

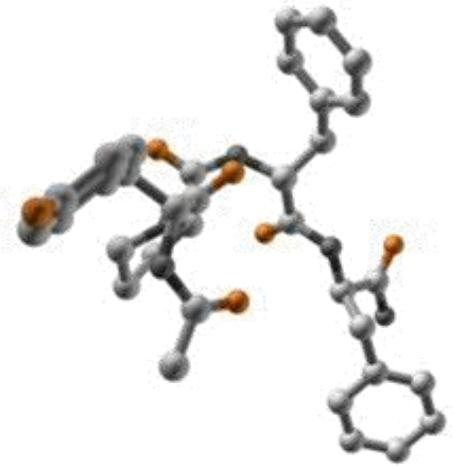
A targeted objective:

To **increase the skin's tolerance threshold of sensitive skin** by decreasing exaggerated nerve stimulation (main cause of sensitive skin type III) for a **soothing effect** by:

- an inhibition of CGRP neuromediator release from sensory neurons,

via

- binding on the μ opioid receptor (proved by an *in vitro* test).



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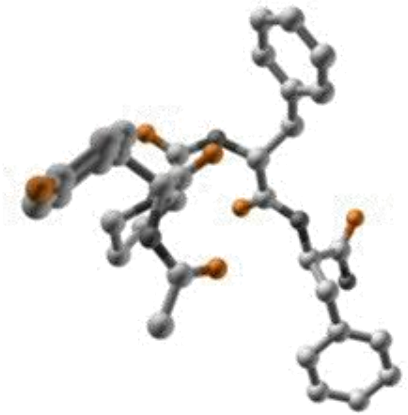
Skinasensyl™

Acetyl Tetrapeptide-15:

Synthetic N-acetylated peptide
N-Acetyl-L-Tyrosyl-L-Prolyl-L-Phenylalanyl-L-Phenylalaninamide
(Acetyl-Tyr-Pro-Phe-Phe-NH_2; Ac-YPFF-NH_2*)

2 versions available:

- a 470 ppm solution, **Skinasensyl™ LS 9749**
- a 1410 ppm powder preservative free, **Skinasensyl™ PW LS 9852**



*** Peptide YPFF-NH_2 is a specific agonist of μ opioid receptor, naturally occurring in the human central nervous system.**

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Efficacy Tests

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Substantiation of Skinasensyl™ Concept

Biological effect on stimulated sensory nerves

Test on sensory neurons
Decrease of CGRP release

↓

Demonstrated mechanism: binding on μ opioid receptor

Test on sensory neurons
Competition with a μ opioid receptor antagonist

↓

Clinical studies:

Cutaneous Capsaicin Sensitivity **Cutaneous Thermal Sensitivity**

In vivo tests on human volunteers
Increase of the skin's tolerance threshold
Soothing effect

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Efficacy On Sensory Neurons Hyper-reactivity

Principle

Sensory neurons in culture have an electrical activity, proportional to the release of neuropeptides as CGRP.

This electrical activity & release of CGRP can be increased by treatment with:

- a stinging agent, capsaicin, acting on vanilloid receptor TRPV1
- a membrane depolarizing agent, KCl.

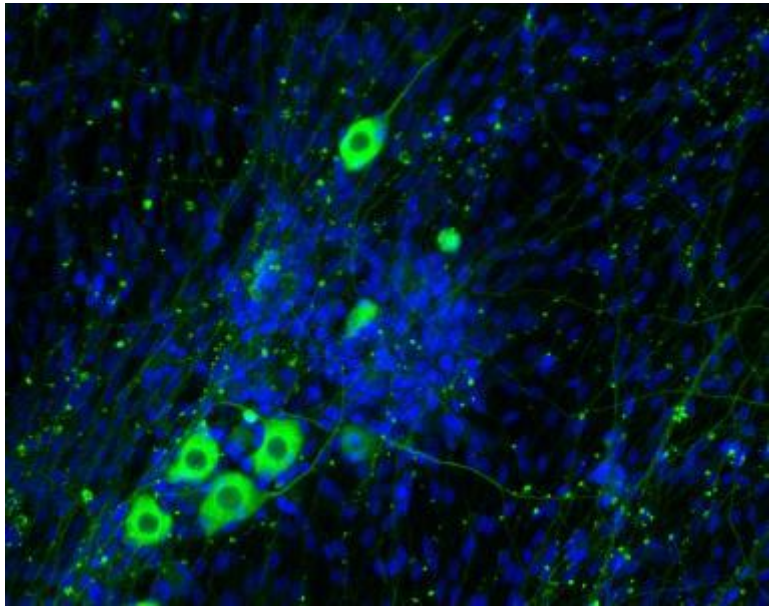
Tested products

- Ac-YPFF-NH_2
- References: Capsazepine: antagonist of vanilloid receptors.
  Verapamil: calcium channels blocker.

Legend:
CGRP = calcitonine gene related peptide

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Culture of Sensory Neurons

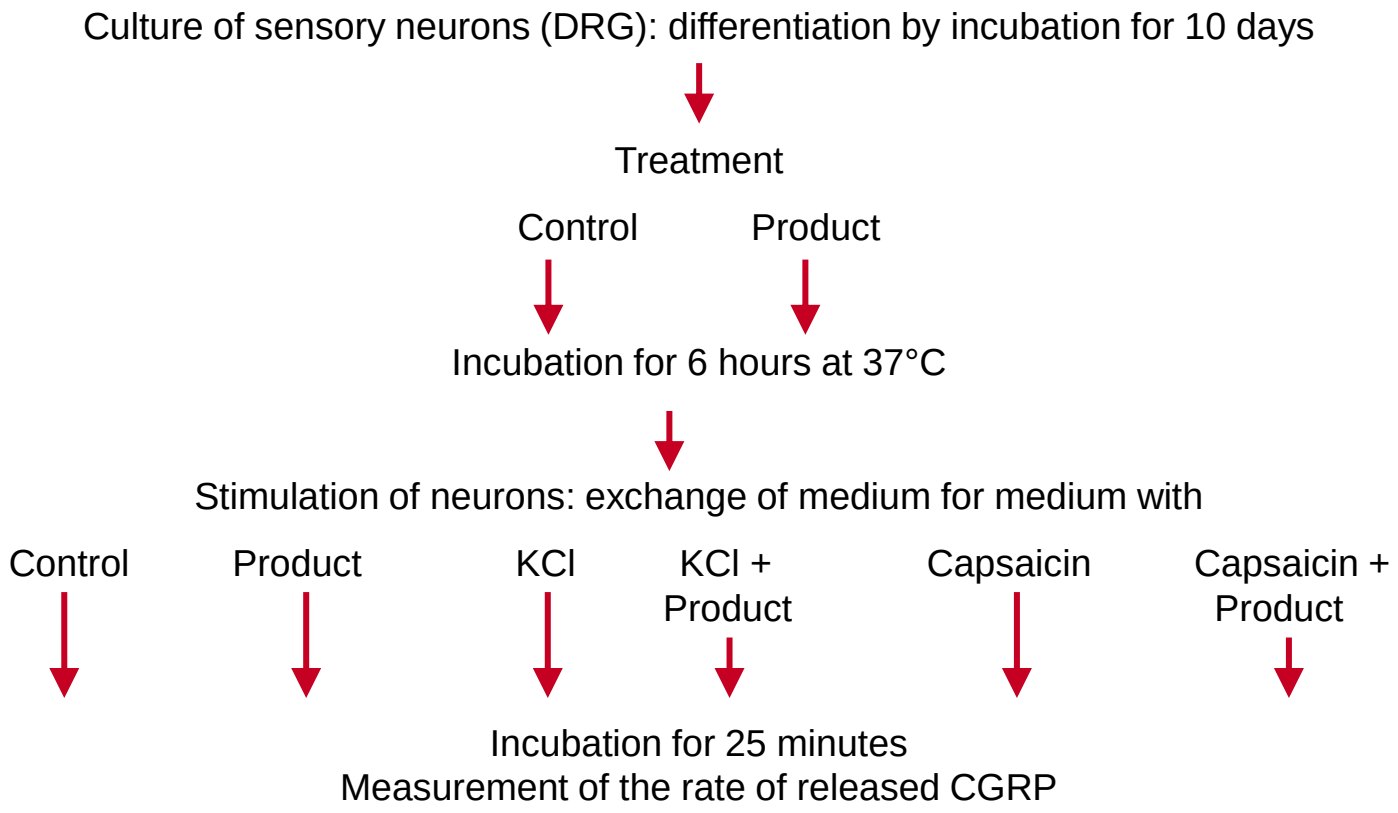


Culture of sensory neurons (Dorsal Root Ganglion (DRG))

(Neuron cells labeled with anti-β-tubulin antibody, nuclei stained with Hoescht solution)

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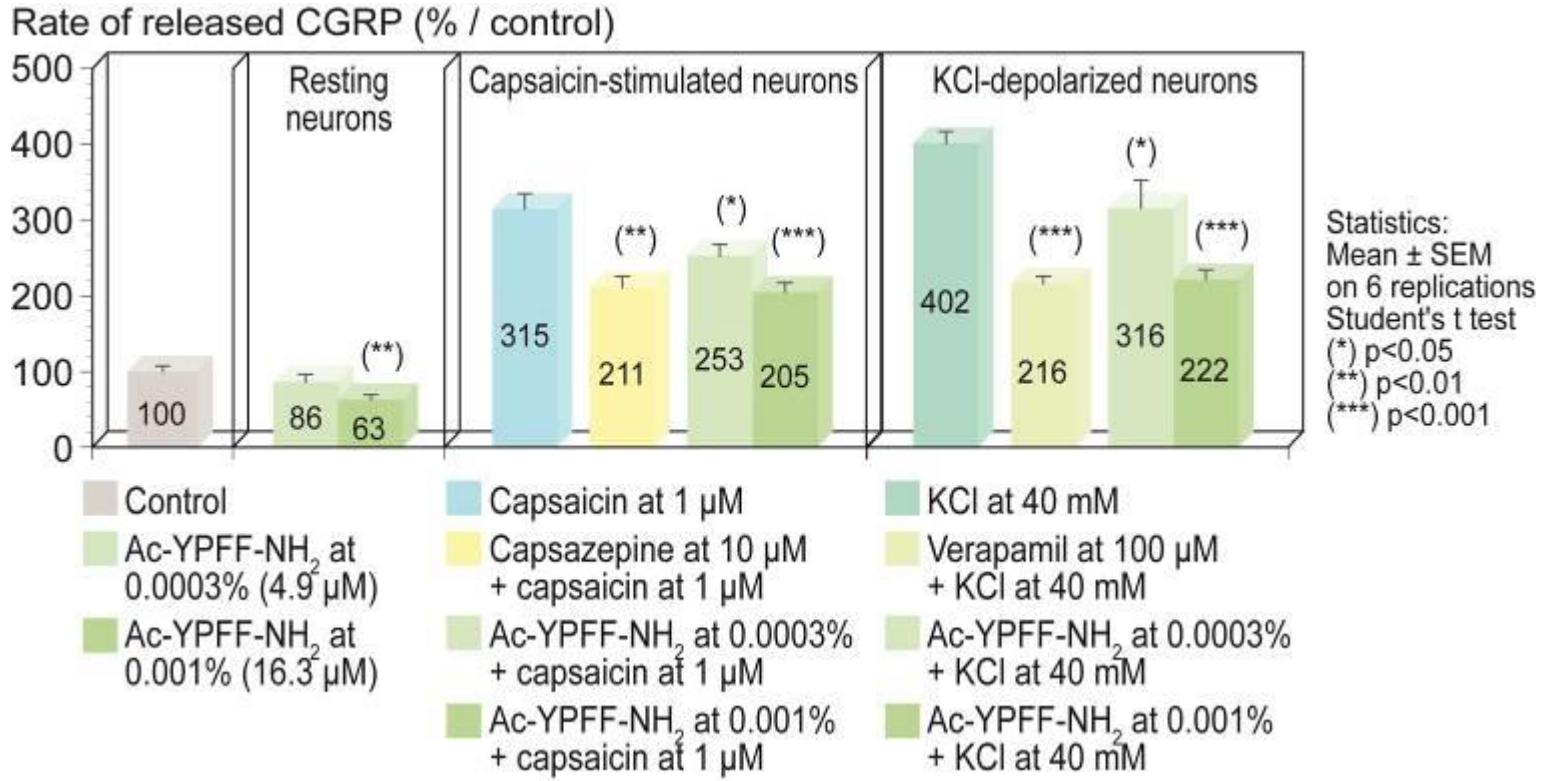
CGRP Release From Cultured Sensory Neurons



Lack of cytotoxicity of products was preliminary tested on cultured sensory neurons by a MTT test.

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CGRP Release From Cultured Sensory Neurons

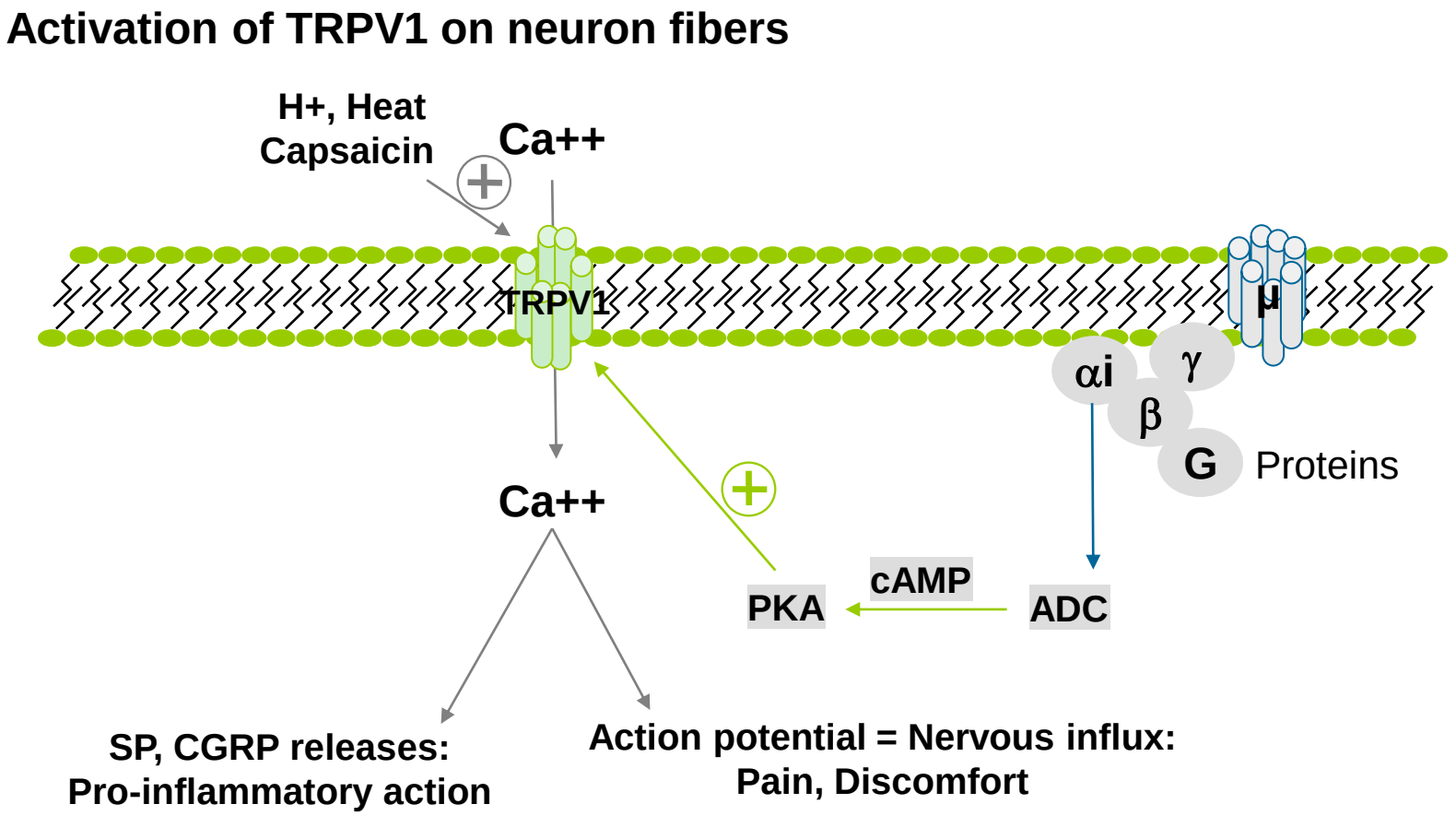


The tetrapeptide Ac-YPFF-NH_2 has significantly reduced the neuronal release of neuromediators.

SKINASENSYL™ has a good potential to appease neuronal hyper-reactivity to environmental stimuli.

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Demonstrated Mechanism Of Action

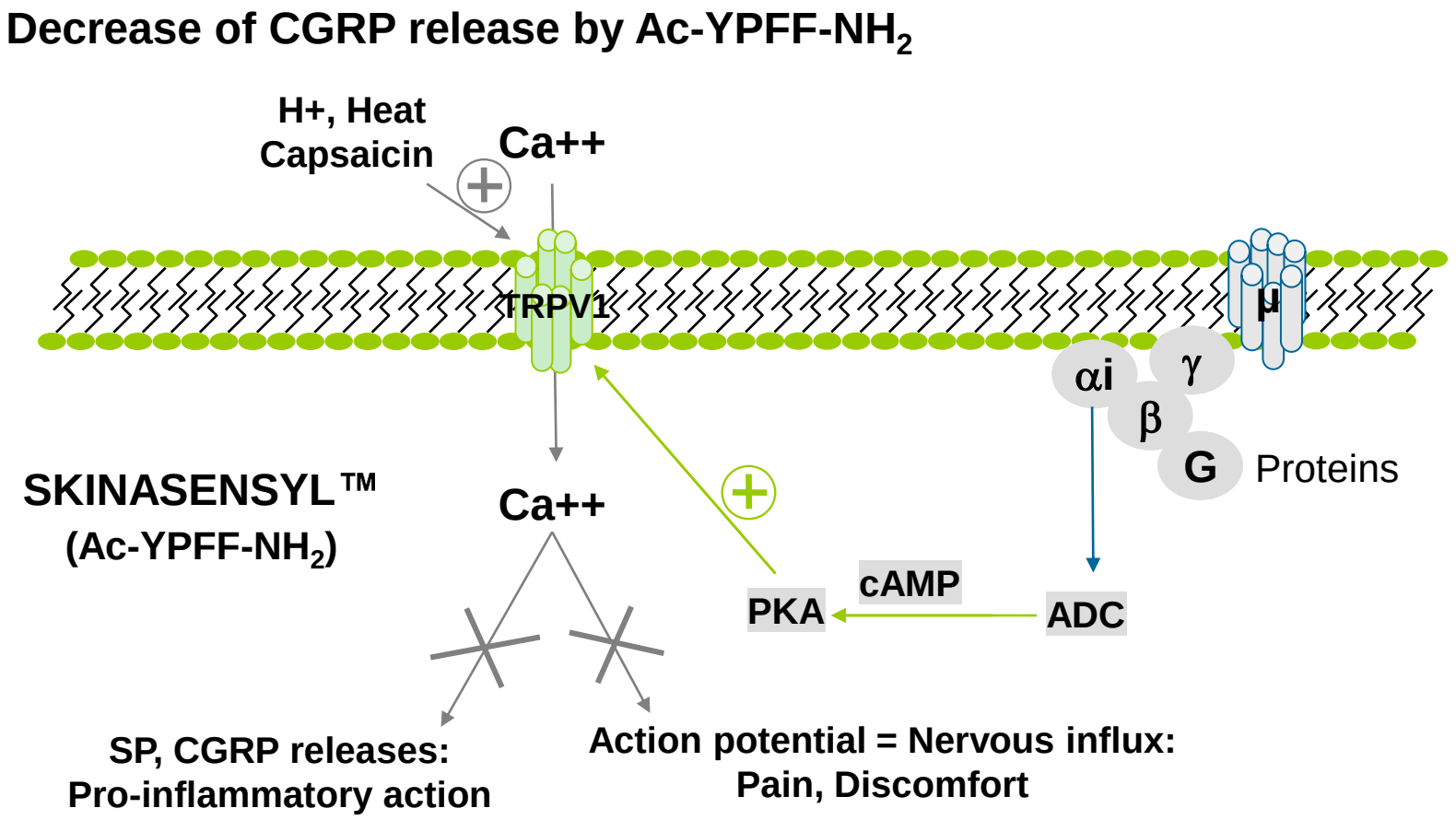


Vetter I *et al*.: The μ opioid agonist morphine modulates potentiation of capsaicin-evoked TRPV1 responses through a cyclic AMP-dependent protein kinase A pathway. Molecular Pain, 2006, 2: 22

Endres-Becker J *et al*.: μ-receptor activation modulate TRPV1 currents in sensory neurons in a model of inflammatory pain. Mol Pharmacol, 2007, 71: 12-8

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Demonstrated Mechanism Of Action



Vetter I *et al*.: The μ opioid agonist morphine modulates potentiation of capsaicin-evoked TRPV1 responses through a cyclic AMP-dependent protein kinase A pathway. Molecular Pain, 2006, 2: 22

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Experimental Validation Of Action of Skinasensyl™ Via μ Receptor

Principle

Competition test between Ac-YPFF-NH_2 and naloxone (opioid receptor antagonist) on the release of CGRP after capsaicin stimulation.

Protocol

Culture of sensory neurons (DRG): differentiation by incubation for 10 days

↓

Treatment

Control | Ac-YPFF-NH_2 | Ac-YPFF-NH_2 + naloxone

↓

Incubation for 6 hours at 37°C

↓

Stimulation of neurons: exchange of medium for medium with

Control | Capsaicin | Capsaicin + Ac-YPFF-NH_2 | Capsaicin + AcYPFF-NH_2 + naloxone

↓

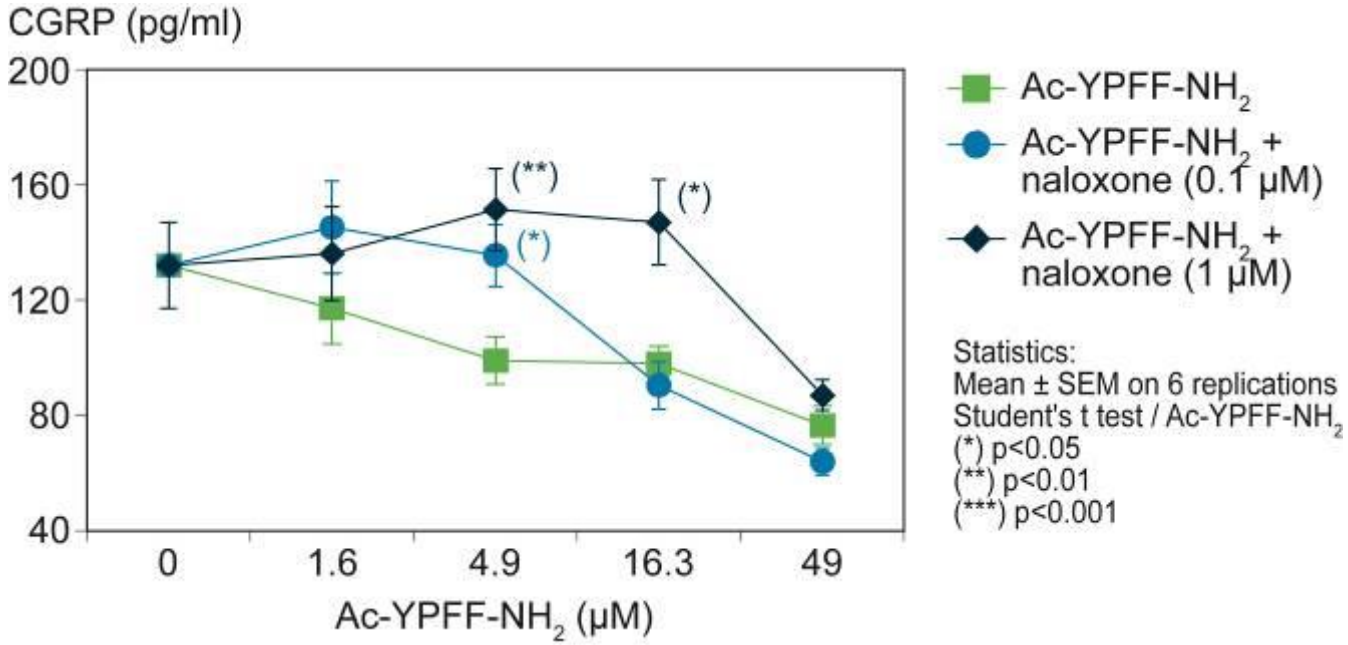
Incubation for 25 minutes

Measurement of the rate of released CGRP

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Experimental Validation Of Action of Skinasensyl™ Via μ Receptor

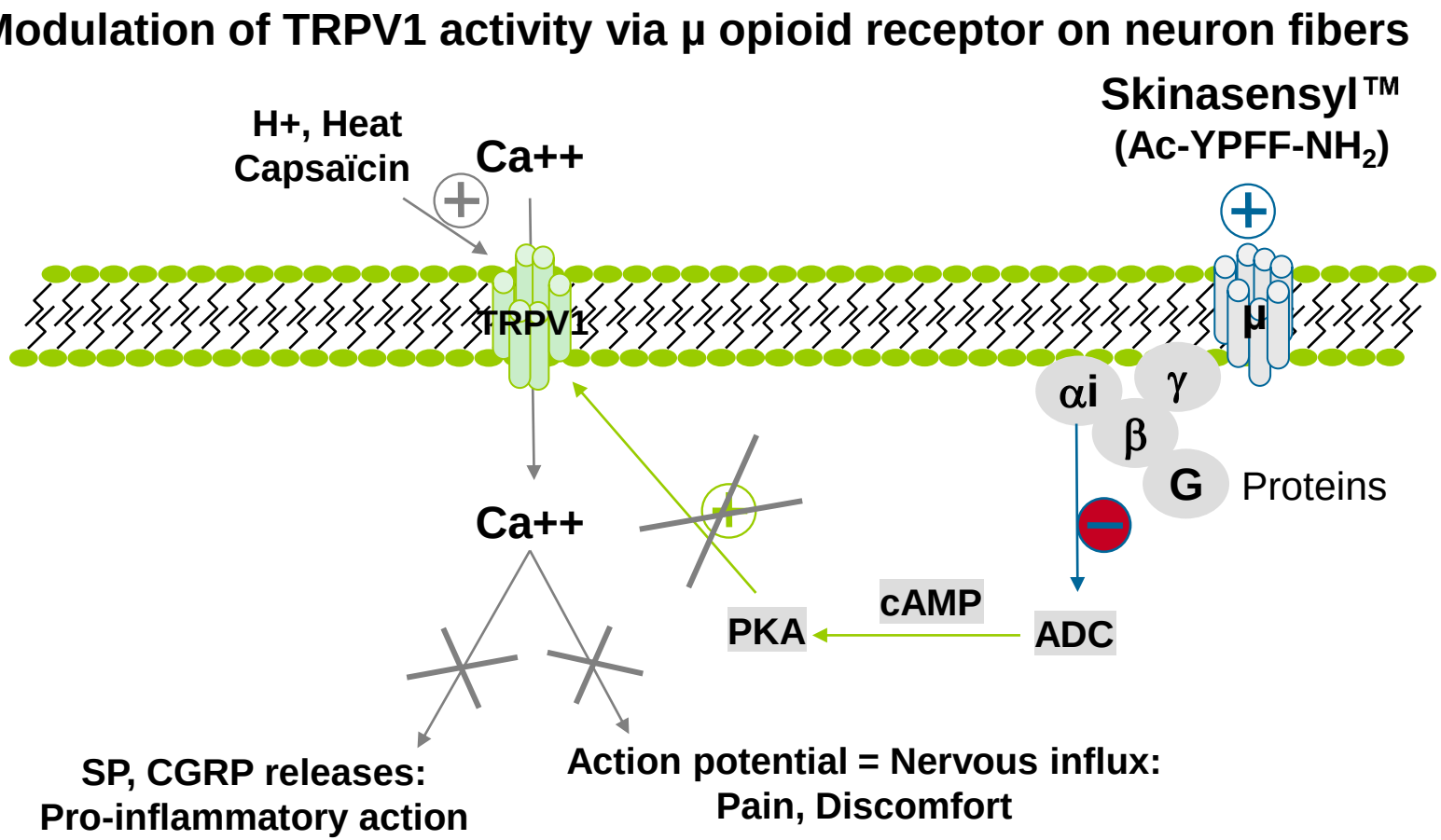
Competitive effect between Ac-YPFF-NH_2 and naloxone after capsaicin stimulation.



As sensory neurons express μ opioid receptor to which naloxone is an antagonist, we can conclude that inhibitory effect of Ac-YPFF-NH_2 on capsaicin-induced release of CGRP is mediated by its binding to μ receptor.

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Demonstrated Mechanism Of Action



Vetter I *et al*.: The μ opioid agonist morphine modulates potentiation of capsaicin-evoked TRPV1 responses through a cyclic AMP-dependent protein kinase A pathway. Molecular Pain, 2006, 2: 22

Endres-Becker J *et al*.: μ-receptor activation modulate TRPV1 currents in sensory neurons in a model of inflammatory pain. Mol Pharmacol, 2007, 71: 12-8

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Clinical Studies

The preventive / soothing effect on peripheric nerve endings can be measured by the level of:

- **Cutaneous Capsaicin Sensitivity,**
- **Cutaneous Thermal Sensitivity (CTS**).

⇨ Target the activity of cutaneous sensory neurons which transmit the itchy, discomfort and pain signals to the Central Nervous System.

⇨ Target skin's tolerance thresholds.

⇨ Do not provoke any alteration of the skin barrier or inflammation.

Cutaneous Capsaicin Sensitivity (Clinical Study)

Principle

Evaluation of individual sensitivity threshold to topically applied capsaicin:

- Application of capsaicin in increasing concentrations
- Stop of the test at the concentration for which discomfort sensations (stinging, itching …) are perceived by the volunteers

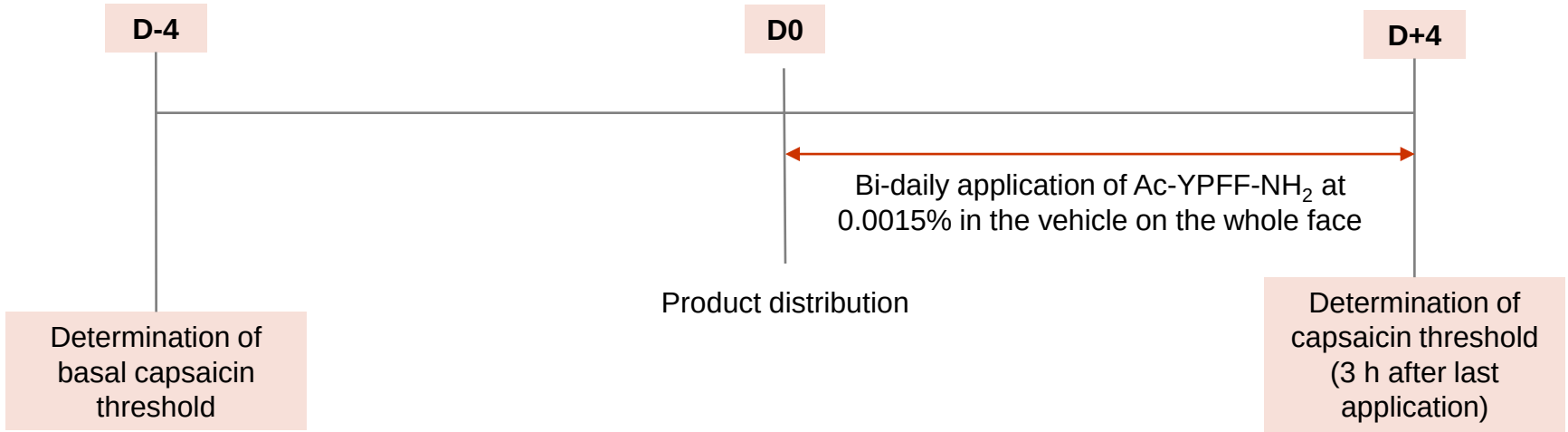
Tested product

Ac-YPFF-NH_2 at 0.0015% in an hydroalcoholic solution
(corresponding to 3% Skinasensyl™ LS 9749)

Preventive / Soothing effect: Δ of capsaicin threshold before and after product application

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Cutaneous Capsaicin Sensitivity (Clinical Study)



Single blind, hemi-face application on the nasolabial folds of 20 human volunteers

Vehicle (hydroalcoholic solution)

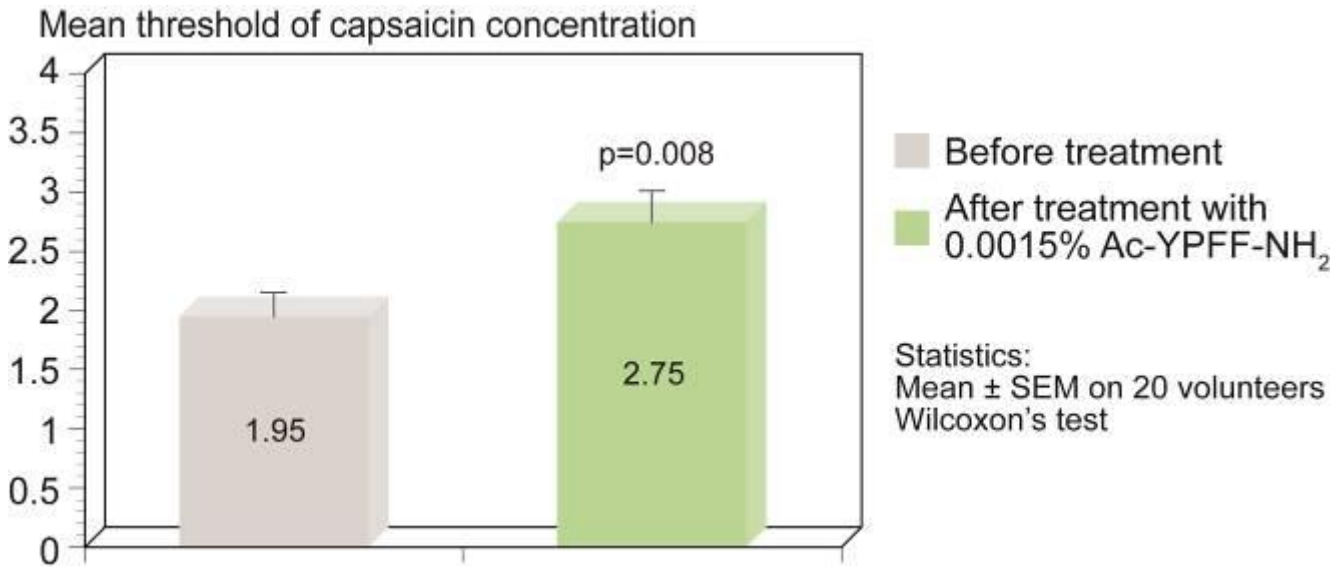


Increasing concentrations of capsaicin in vehicle, noted from 1 to 5

Determination of capsaicin concentration inducing a discomfort sensation

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Cutaneous Capsaicin Sensitivity (Clinical Study)



Δ of capsaicin threshold before and after Ac-YPFF-NH_2 application: 0.80 (p 0.008). During the treatment the skin tolerance threshold is increased. Thanks to SKINASENSYL the skin is able to tolerate higher concentrations of capsaicin.

⇨ **preventive soothing effect**

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Cutaneous Capsaicin Sensitivity (Clinical Study)

Principle

Evaluation of individual discomfort and pain threshold to temperature :

- Application of a thermal probe
- Report of discomfort and pain sensations by volunteer

Tested product

Emulsion with 2% Skinasensyl™ (Ac-YPFF-NH_2 at 0.001%)

Placebo emulsion

Soothing effect: Δ of temperature threshold before and after product application

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Cutaneous Capsaicin Sensitivity (Clinical Study)

16 human volunteers
Single randomized application on the back for an immediate effect

Placebo emulsion

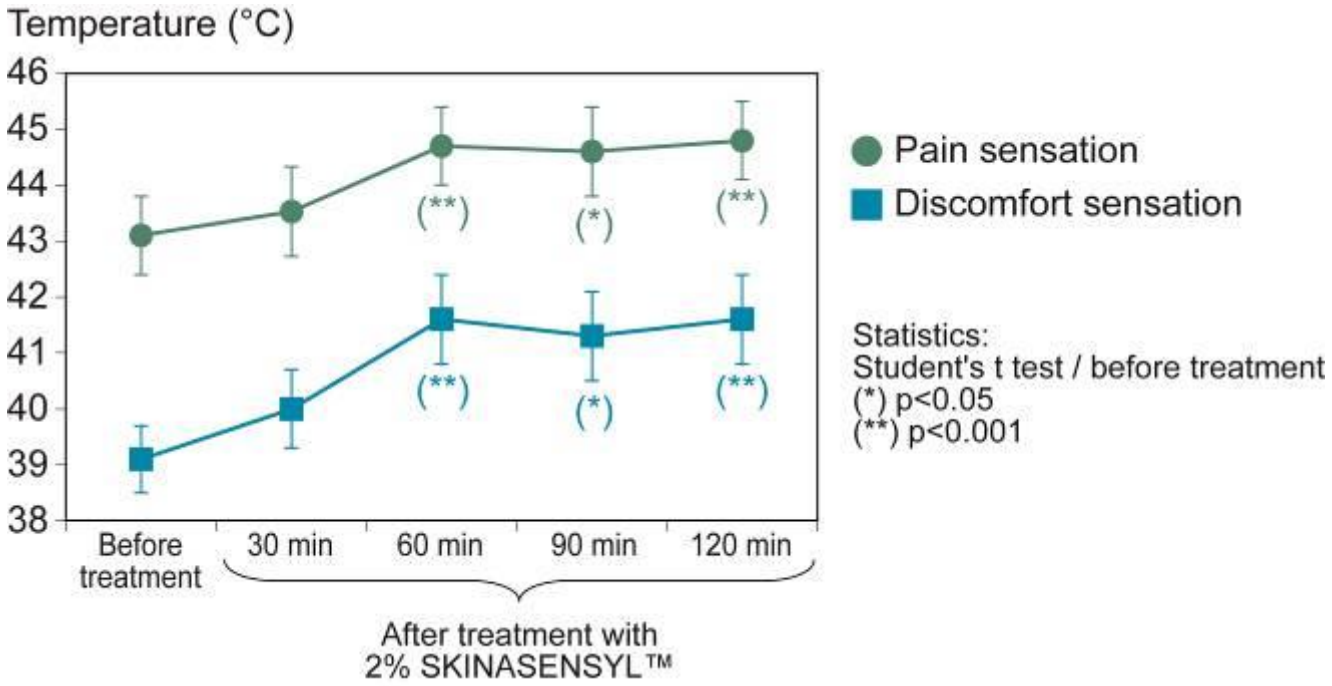


Emulsion with 2% SKINASENSYL™ (i.e. 0.001% Ac-YPFF-NH_2)

Evaluation of discomfort and pain sensations, using a thermal probe before and up to 120 minutes after application

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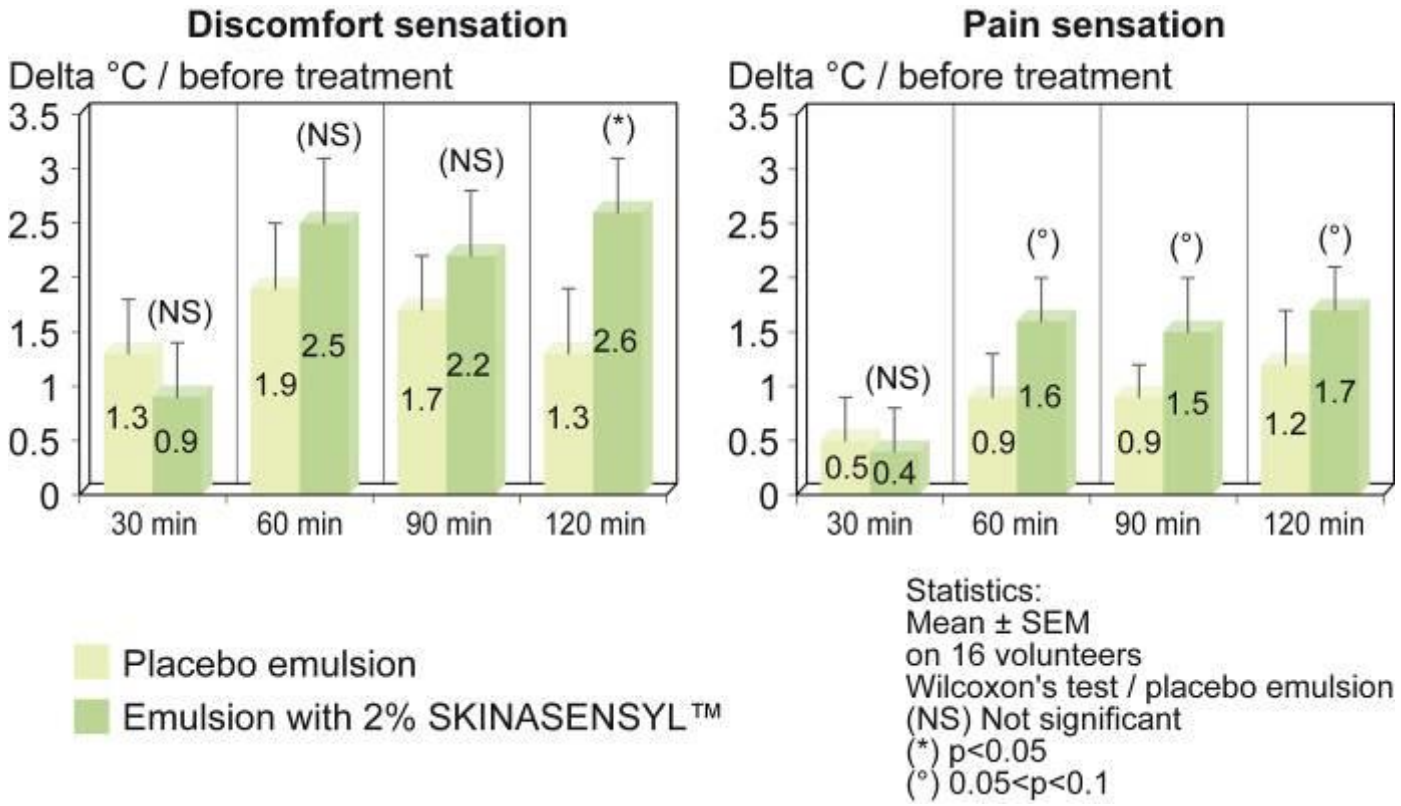
Cutaneous Capsaicin Sensitivity (Clinical Study)



Perceptible increase threshold for sensorial discomfort and painful (significant 60 min after application)
Soothing effect maximum 60 minutes after application, stable after.

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Cutaneous Capsaicin Sensitivity (Clinical Study)



Soothing effect is significant *vs* placebo 120 minutes after application of the emulsion (with a tendency 60 and 90 minutes after application).

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Conclusion

Significant and demonstrated reduction of skin hyper-reactivity (*in vitro* & *in vivo* tests)**:**

- **Decrease of nerve stimulation**

  Protection against neurogenic hyper-reactivity

  ↘ CGRP release from sensory neurons
  via binding on the μ opioid receptor

- **Increase of skin tolerance threshold**

  Soothing effect for over-reactive skin

  ↗ tolerance threshold to
  - capsaicin: stinging, burning, itching sensations
  - temperature: sensorial discomfort, pain

This prevention can be extended to other stimuli also known to induce neuron stimulation.

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Skinasensyl™ LS 9749
Technical Data

Aspect: colorless clear liquid, of weak odor

Dose of use: 1 - 3%

Solubility: soluble in water, insoluble in oils.

INCI name: Water (and) Glycerin (and) Coco-Glucoside (and) Acetyl Tetrapeptide-15

Preservative: sorbic acid.

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Skinasensyl™ LS 9749
Technical Data

Aspect: white powder, of weak odor.

Dose of use: 0.3 - 1%.

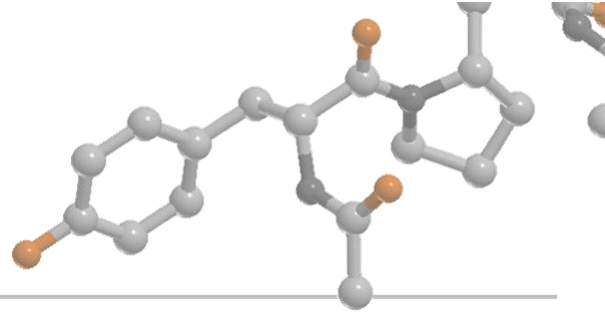
Solubility: soluble in water, insoluble in oils.

INCI name : Mannitol (and) Sodium Citrate (and) Acetyl Tetrapeptide-15

Preservative: preservative free.

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The Efficient Solution for Neurosensitive skin



SKINASENSYL™ is a new generation of **soothing active ingredient** allowing to offer strong and perceivable benefits to sensitive skin.

Thanks to **SKINASENSYL™** the tolerance threshold of the most sensitive skin is increased.

Over-reactive skin feels soothed:

- it recovers the right to comfort,
- it recovers the right to high performing skin care products.

A soothing effect on the skin…

Who says that sensitive skin is an inevitable fate?

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